

58 :

```
syms t T1;
f1=sin(t)*dirac(t-T1/2);
f1_result=int(f1,t,-inf,inf);%第一题
disp(f1_result);
f2=(t^3+t+2)*dirac(t-1);
f2_result=int(f2,t,-inf,inf);%第二题
disp(f2_result);
```

输出:

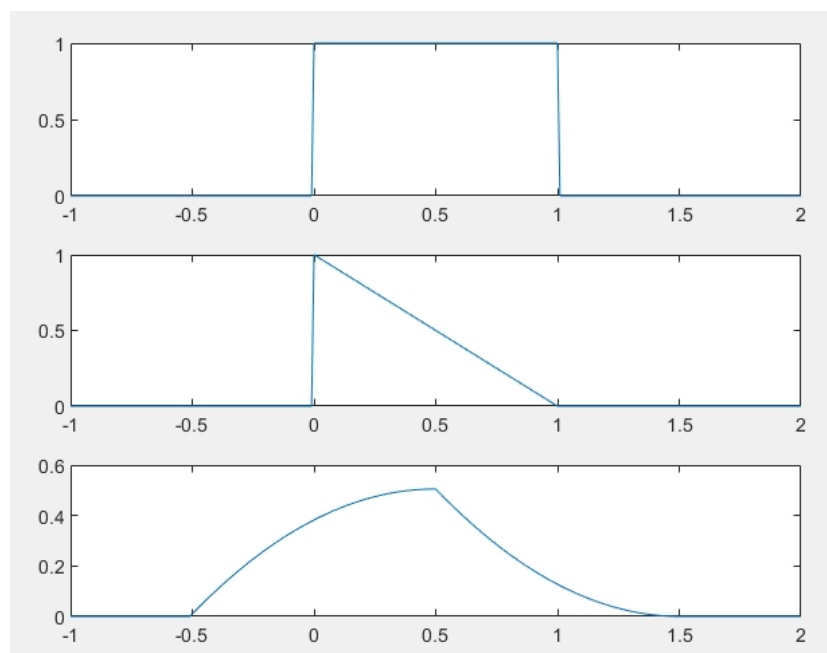
```
sin(T1/2)
```

```
4
```

附加题:

```
t=-1:0.01:2;
f3=(t>=0&t<=1);
f4=(t>= 0&t<=1).*(1 - t);
y=conv(f3,f4,'same')*0.01;
figure;
subplot(3,1,1);
plot(t,f3);
subplot(3,1,2);
plot(t,f4);
subplot(3,1,3);
plot(t,y);
```

输出



最下面图象为卷积的结果。